UCC103 – PRINCIPLES OF COMPUTING
Chapter 5: Computer Number Systems

5.1 Introduction to Digital Systems

Electronic circuits/systems are classified as being analogue or digital, the distinction between the two circuits is not the semi-circular material used to construct them but rather the way they are operating, that is Current and voltage variations during performance. Analog electronics deals with things that are continuous in nature and digital electronics deals with things that are discrete in nature. But they are very much interlinked. By contrast, the digital circuit is the one which voltage levels alternate among a finite number of distance value that is, they are 2 voltage levels high a low.

Digital circuits produce discrete outputs, distinct circuits have individual components e.g. resistors, and transistors. Hybrids circuits contain both integrated and discrete components. Digital circuits are often called logic circulatory because the level of each output voltage depends on several input voltages and the inputs voltage may appear in many different combinations.

5.2 Number Systems.

Number System- A number system defines a set of values to represent quantity. We talk about the number of people attending a class, the number of modules taken by each student and use numbers to represent grade.

Number System can be categorized in two systems:-

a) **Non-Positional Number System**- In ancient times, people used to count on fingers, when the fingers became insufficient for counting, stones, pebbles or sticks were used to indicate values. But it was very difficult to perform arithmetic with such a number system as there is no symbol for zero.
b) **Positional Number System**- In this system the value of each digit is defined not by the symbol but also by the symbol position. Positional Number System is used to perform arithmetic. Existing Positional number system is decimal number system. Apart from the decimal number system, there are binary number system, octal number system and hexadecimal number system.

The base of a number system is indicated by a subscript (decimal number) and this will be followed by the value of the number. For example $(952)_{10}$, $(456)_{8}$, $(314)_{16}$

Number System that are used by the computers-

- Decimal System
- Binary System
- Octal System
- Hexadecimal System

Numbers are difficult to define, they are simply a symbolic representation of ideas; number systems are positional in nature and therefore a symbolic of numbers has weights.

5.3. Decimal Number System

The decimal system is the system which we use in everyday counting. The number system includes the ten digits from 0 through 9 (0 – 9). These digits are recognized as the symbols of the decimal system and are thus called decimal digits. The position of each digital has a numerical weight and each digital is a multiplier of the weight of its position e.g. 10^2, 10^1 ,10^0 – weights. When our count exceeds the highest digit available, the next digit to the left is incremented and the original digit is reset to zero. For example:

$9_{10} + 1_{10} = 10_{10}$

Each digit in a base ten number represents units ten times the units of the digit to its right. For example-

$9542_{10} = 9000 + 500 + 40 + 2 = (9 \times 10^3) + (5 \times 10^2) + (4 \times 10^1) + (2 \times 10^0)$

$130_{10} = 100 + 30 + 0 = (1x10^2) + (3x10^1) + (0x10^0)$

Because we are dealing with a base-10 system, each digit to the left of another digit is weighted ten times higher. Using exponential notation, we can imagine the number 10 as representing:

5.4. Binary Number Systems

Computers do not use the decimal system for counting and arithmetic. Their CPU and memory are made up of millions of tiny switches that can be either in ON and OFF states. 0 represents OFF and 1 represents ON. The number system includes the two digits from 0 and 1(0,1). Counting Similar to decimals, binary digits have a positional weight. Each bit is weighted twice as much as the bit to the right of it

5.5. Octal Number System

The octal system is commonly used with computers. The name octal implies eight; octal number system has a radix of eight and uses the following 8 digit 0,1,2,3,4,5,6, and 7 has base 8. The octal system uses a power of 8 to determine the digit of a number's position.

5.6. Hexadecimal System

Hexadecimal System- Hexadecimal is another number system that works exactly like the decimal, binary and octal number systems, except that the base is 16. Each hexadecimal represents a power of 16. The system uses 0 to 9 numbers and A to F characters to represent 10 to 15 respectively.

5.7. Number Systems Conversions

Any number in one number system can be converted into any other number system. There are the various methods that are used in converting numbers from one base to another.

Decimal Number	Binary Number	Octal Number	Hexadecimal Number
0	0000	0	0
1	0001	1	1
2	0010	2	2
3	0011	3	3
4	0100	4	4
5	0101	5	5
6	0110	6	6
7	0111	7	7
8	1000		8
9	1001		9
10	1010		A
11	1011		B
12	1100		C
13	1101		D
14	1110		E
15	1111		F

5.8. Conversions of Decimal to other Number System

a). Decimal to Binary- The method that is used for converting of decimals into binary is known as the remainder method. We use the following steps in getting the binary number-

1) Divide the decimal number by 2.
2) Write the remainder (which is either 0 or 1) at the right most position.
3) Repeat the process of dividing by 2 until the quotient is 0 and keep writing the remainder after each step of division.
4) Write the remainders in reverse order.

Example- Convert $(45)_{10}$ into binary number system.

		Remainder
2	45	1
2	22	0
2	11	1
2	5	1
2	2	0
2	1	1
	0	

Thus $(45)_{10} = (101101)_2$

Note- In every number system-

- **The first bit from the right is referred as LSB (Least Significant Bit)**
- **The first bit from the left is referred as MSB (Most Significant Bit)**

b). Conversion of Decimal to Octal- In converting decimal to octal, we follow the same process of converting decimal to binary. Instead of dividing the number by 2, we divide the number by 8.

Example- Convert $(45)_{10}$ into octal number system.

		Remainder
8	45	5 ↑
8	5	5 |
8	0	

Thus $(45)_{10} = (55)_8$.

c). Conversion of Decimal to Hexadecimal- We divide by 16 instead of 2 or 8. If the remainder is in between 10 to 16, then the number is represented by A to F respectively.

Example- Convert $(45)_{10}$ into hexadecimal.

		Remainder
16	45	D ↑
16	2	2 |
	0	

Thus $(45)_{10} = (2D)_{16}$.

5.8.1. Conversion of Decimal Fractions

Conversions of Decimal Fractions to Binary Fractions- For converting decimal fractions into binary fractions, we use multiplication. Instead of looking for a remainder we look for an integer. The following steps are used in getting the binary fractions-

1) Multiply the decimal fraction by 2.
2) If a non-zero integer is generated, record the non-zero integer otherwise record 0.
3) Remove the non-zero integer and repeat the above steps till the fraction value becomes 0.
4) Write down the number according to the occurrence.

Example- Find the binary equivalent of $(0.75)_{10}$.

Number (to be recorded)

$0.75 \times 2 = 1.50$ 1

$0.50 \times 2 = 1.00$ 1

Thus $(0.75)_{10} = (0.11)_2$.

Moreover, we can write $(45.75)_{10} = (101101.11)_2$.

Remark- If the conversion is not ended and still continuing; we write the approximation in 16 bits.

Example- Find the binary equivalent of $(0.9)_{10}$.

Number (to be recorded)

$0.9 \times 2 = 1.8$ 1

0.8 × 2 = 1.6 1
0.6 × 2 = 1.2 1
0.2 × 2 = 0.4 0
0.4 × 2 = 0.8 0
0.8 × 2 = 1.6 1
0.6 × 2 = 1.2 1
0.2 × 2 = 0.4 0
0.4 × 2 = 0.8 0
0.8 × 2 = 1.6 1
0.6 × 2 = 1.2 1
0.2 × 2 = 0.4 0
0.4 × 2 = 0.8 0
0.8 × 2 = 1.6 1
0.6 × 2 = 1.2 1
0.2 × 2 = 0.4 0
0.4 × 2 = 0.8 0
0.8 × 2 = 1.6 1
Thus $(0.9)_{10} = (0.1110011001100110011001)_2$.

Conversions of Decimal Fractions to Octal Fractions –We follow the same steps of conversions of decimal fractions to binary fractions. Here we multiply the fraction by 8 instead of 2.
Example- Find the octal equivalent of $(0.75)_{10.}$
Number (to be recorded)
0.75 × 8 = 6.00 6

Thus $(0.75)10= (0.6)_8$.
And $(45.75)10= (55.6)_8$.

Conversions of Decimal Fractions to Hexadecimal Fractions – Here we multiply the fraction by 16 instead of 2 or 8. If the non-zero integer is in between 10 to 16, then the number is represented by A to F respectively.

Example- Find the hexadecimal equivalent of $(0.75)_{10}$.
Number (to be recorded)
0.75 × 16 = 12.00C (12 = C)

Thus $(0.75)10= (0.C)_{16}$.
And $(45.75)10= (2D.C)_{16.}$

5.9. Conversions of Binary to Other Numbers

5.9.1. Conversions of Binary to Decimal

In converting binary to decimal, we use the following steps-

1) Write the weight of each bit.
2) Get the weighted value by multiplying the weighted position with the respective bit.
3) Add all the weighted value to get the decimal number.

Example- Convert $(101101)_2$ into decimal number system.

Binary Number	1	0	1	1	0	1
Wt. of Each bit	2^5	2^4	2^3	2^2	2^1	2^0
Weighted Value	1×2^5	0×2^4	1×2^3	1×2^2	0×2^1	1×2^0
Solved Multiplication	32	0	8	4	0	1

Thus $(101101)_2 = (1 \times 2^{5)} + (0 \times 2^{4)} + (1 \times 2^3) + (1 \times 2^2) + (0 \times 2^1) + (1 \times 2^0)$.
$= 32 + 0 + 8 + 4 + 0 + 1$
$= 45_{10}$

5.9.1.1. Conversions of Binary Fractions to Decimal Fractions

The conversions of binary fractions to the decimal fractions is similar to conversion of binary numbers to decimal numbers. Here, instead of a decimal point we have a binary point. The exponential expressions (or weight of the bits) of each fractional placeholder is 2^{-1}, 2^{-2}……

Example- Convert $(101101.11)_2$ into decimal number system.

Binary Number	1	0	1	1	0	1	1	1
Wt. of Each bit	2^5	2^4	2^3	2^2	2^1	2^0	2^{-1}	2^{-2}
Weighted Value	1×2^5	0×2^4	1×2^3	1×2^2	0×2^1	1×2^0	1×2^{-1}	1×2^{-2}
Solved Multiplication	32	0	8	4	0	1	0.5	0.25

Thus $(101101.11)_2 = (1 \times 2^5) + (0 \times 2^4) + (1 \times 2^3) + (1 \times 2^2) + 0 \times 2^1 + (1 \times 2^0) + (1 \times 2^{-1}) + (1 \times 2^{-2})$

$= 32 + 0 + 8 + 4 + 0 + 1 + 0.5 + 0.25 = 45.75_{10}$

5.9.2. Conversions of Binary to Octal

We use the following steps in converting binary to octal-

1) Break the number into 3-bit sections starting from LSB to MSB.
2) If we do not have sufficient bits in grouping of 3-bits, we add zeros to the left of MSB so that all the groups have proper 3-bit number.
3) Write the 3-bit binary number to its octal equivalent.

Example- Convert $(101101)_2$ into octal.

Binary Number	101101	
3-bit Binary Number sections	101	101
Octal Number	5	5

Thus $(101101)_2 = (55)_8$.

Example- Convert $(1101101)_2$ into octal.

Binary Number	001101101		
3-bit Binary Number Sections	001	101	101
Octal Number	1	5	5

Thus $(1101101)_2 = (155)_8$.

5.9.2.1. Conversion of Binary Fractions to Octal Fractions

We use the following steps in converting binary fractions to octal fractions-

1) Break the fraction into 3-bit sections starting from LSB to MSB.
2) In order to get a complete grouping of 3 bits, we add trailing zeros in MSB.
3) Write the 3-bit binary number to its octal equivalent.

Example- Convert $(101101.11)_2$ into octal.

Binary Number	101	101	110
Octal Number	5	5	6

Thus $(101101)2 = (55.6)_8$.

5.9.3. Conversion of Binary to Hexadecimal

We convert binary to hexadecimal in the similar manner as we have converted binary to octal. The only difference is that here, we form the group of 4 bits.

Example- Convert $(101101)_2$ into hexadecimal.

Binary Number	0010	1101
Hexadecimal Number	2	D

Thus $(101101)2 = (2D)_{16}$.

5.9.3.1. Conversion of Binary Fractions to Hexadecimal Fractions

We convert binary fractions to hexadecimal fractions in the similar manner as we have converted binary fractions to octal fractions. The only difference is that here we form the group of 4 bits.

Example- Convert $(101101.11)_2$ into hexadecimal.

Binary Number	0010	1101	1100
Decimal Number	2	13	12
Hexadecimal Number	2	D	C

Thus $(101101.11)_2 = (2D.C)_{16}$.

5.10. Conversions of Octal to Other Numbers

5.10.1. Conversions of Octal to Decimal

We follow the same steps of conversion of binary to decimal. The only difference is that here weight of nth bit is 8^{n-1} instead of 2^{n-1}.

Example- Convert (55)8 into decimal number system.

Octal Number	5	5
Wt. of Each bit	8^1	8^0
Weighted Value	5×8^1	5×8^0
Solved Multiplication	40	5

Thus $(55)_8 = 40 + 5.$
$= 45_{10}$

5.10.1.1. Conversions of Octal Fractions to Decimal Fractions-

The weight of the bit of the fraction placeholder is 8^{-1}, 8^{-2} …We follow the same steps of conversion of binary fractions to decimal fractions.

Example- Convert $(55.6)_8$ into decimal number system.

Octal Number	5	5	6
Wt. of Each bit	8^1	8^0	8^{-1}
Weighted Value	5×8^1	5×8^0	6×8^{-1}
Solved Multiplication	40	5	0.75

Thus $(55.6)_8 = 40 + 5 + 0.75 = 45.75_{10}$

5.10.2. Conversions of Octal to Binary-

We use the following steps in converting octal to binary-

1) Convert each octal digit into 3-bit binary equivalent.
2) Combine the 3-bit section by removing the spaces to get the binary number.

Example- Convert $(55)_8$ into binary.

Octal Number	5	5
3-bit Binary Number	101	101
Binary Number	101101_2	

Thus $(55)_8 = (101101)_2$.

Example- Convert $(456)_8$ into binary.

Octal Number	4	5	6
3-bit Binary Number	100	101	110
Binary Number	100101110_2		

Thus $(456)_8 = (100101110)_2$.

5.10.2.1. Conversion of Octal Fractions to Binary Fractions

We follow the same steps of conversion of octal to binary.

Example- Convert $(55.6)_8$ into binary.

Octal Number	5	5	6
Binary Number	101	101	110

Thus $(55.6)_8 = (101101.11)_2$.

5.10.3. Conversion of Octal to Hexadecimal

The conversion involves the following steps-

1) Convert each octal digit to 3–bit binary form.
2) Combine all the 3-bit binary numbers.
3) Group them in 4-bit binary form by starting from LSB to MSB.
4) Convert these 4-bit blocks into their hexadecimal symbols.

Example- Convert $(55)_8$ into hexadecimal.

Octal Number	5	5
3-bit Binary Number sections	101	101
Combine the 3-bit sections	101101_2	

Split number into 4-bit Binary Number sections	0010	1101
Hexadecimal Number	2	D

Thus $(55)_8 = (2D)_{16}$.

5.10.3.1. Conversions of Octal Fractions to Hexadecimal Fractions

The method of conversion is based on the same procedure that we have discussed in conversions of octal to hexadecimal.

Example- Convert $(55.6)_8$ into hexadecimal.

Octal Number	5	5	6
Binary Number	101	101	110

1) Combining the 3-bit binary block, we have 101101.110.
2) Grouping them in 4 bit binary form (ignoring the decimal point)

Binary Number	0010	1101	1100
Hexadecimal Number	2	D	C

Thus $(55)8 = (2D.C)_{16}$.

5.11. Conversions of Hexadecimal to Other Numbering systems

5.11.1 Conversions of Hexadecimal to Decimal

We do the conversion of hexadecimal to decimal as we have done the conversion of binary to decimal. Here weight of nth bit is 16^{n-1} instead of 2^{n-1}.

Example- Convert $(2D)_{16}$ into decimal.

Hexadecimal Number	2	D (=13)
Wt. of Each bit	16^1	16^0
Weighted Value	2×16^1	13×16^0
Solved Multiplication	32	13

Thus $(2D)_{16} = 32 + 13 = 45_{10}$.

5.11.1.1 Conversions of Hexadecimal Fractions to Decimal Fractions

We do the conversion of hexadecimal fractions to decimal fractions in the similar manner as we have done the conversion of binary fractions to decimal fractions. Here weight of bit is 16^{-1}, 16^{-2} …….

Example- Convert $(2D.C)_{16}$ into decimal.

Hexadecimal Number	2	D (13)	C (12)
Wt. of Each bit	16^1	16^0	16^{-1}
Weighted Value	2×16^1	13×16^0	12×16^{-1}
Solved Multiplication	32	13	0.75

Thus $(2D.C)_{16} = 32 + 13 + 0.75 = 45.75_{10}$.

5.11.2. Conversions of Hexadecimal to Binary

We use the following steps-

1) Convert each hexadecimal digit to its 4-bit binary equivalent.
2) Combine all the binary numbers.

Example- Convert $(2D)_{16}$ into binary.

Hexadecimal Number	2	D
Binary Number	0010	1101

Thus $(2D)16 = (00101101)_2 = (101101)_2$.

5.11.2.1. Conversions of Hexadecimal Fractions to Binary Fractions

We use the same steps of hexadecimal to binary conversion.

Example- Convert $(2D.C)_{16}$ into binary.

Hexadecimal Number	2	D	C	
Binary Number	0010	1101	1100	

Thus $(2D)_{16} = (00101101.1100)_2 = (101101.11)_2$.

5.11.3. Conversions of Hexadecimal to Octal

We convert each hexadecimal digit in binary. Combine all the binary numbers. Again group them into 3-bit form. Convert the 3-bit block in octal.

Example- Convert $(2D)_{16}$ into octal.

Hexadecimal Number	2	D
4-bit Binary Number	0010	1101
Combined Number	00101101	

3-bit Binary Number groups	000	101	101
Octal Number	0	5	5

Thus $(2D)_{16} = (55)_8$.

5.11.3.1. Conversions of Hexadecimal Fractions to Octal Fractions

We follow the same steps of hexadecimal to octal conversion.

Example- Convert $(2D.C)_{16}$ into octal.

Hexadecimal Number	2	D (=13)	C (=12)
Binary Number	0010	1101	1100

Combining the binary number, we get 00101101.1100=101101.11

Grouping the binary number in 3-bit

Binary Number	101	101	110
Octal Number	5	5	6

Thus $(2D.C)_{16} = (55.6)_8$.

5.12. Revision Questions

Perform the following Number systems conversions

i. 512_{10} to Binary
ii. 5467_8 to Hexadecimal
iii. 10100110_2 to Octal
iv. $7ADF3C_{16}$ to Octal
v. 100100111010_2 to Hexadecimal
vi. $CB235_{16}$ to Decimal